

ENTHUSIAST ADVANCE COURSE

LEADER PHASE I & MEPS-I (A) PHASE II To TARGET NEET 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 18-06-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	3	1	1	4	3	1	2	2	2	2	2	1	3	2	1	2	4	2	2	1	3	2	3	1	3	2	1	3	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	2	3	1	4	2	2	3	2	2	1	2	2	2	3	3	1	1	3	1	2	2	3	3	2	2	2	2	3	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	4	2	3	2	2	4	4	1	4	3	2	1	3	3	1	3	2	1	4	3	2	4	1	4	4	4	2	2	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	2	2	4	2	2	1	1	4	4	2	4	4	2	2	2	4	3	3	2	2	2	4	4	3	4	2	3	4	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	4	3	3	1	3	4	2	2	1	3	1	4	4	4	3	1	4	1	1	3	4	3	2	2	2	2	4	2	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	1	1	2	3	3	1	4	2	3	4	2	3	4	1	4	3	4	2	4	3	2	2	3	4	2	4	4	3	3
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	2	3	3	2	1	3	2	4	1	1	1	2	2	2	4	2	2	4	3	4										

HINT - SHEET

SUBJECT : PHYSICS

SECTION-A

1. Ans (1)

$$f_1 \lambda_1 = f_2 \lambda_2$$

$$(300)(1) = (f_2)(1.5)$$

$$200 \text{ Hz} = f_2$$

2. Ans (3)

Maximum particle velocity $v_{\max} = \omega a$ and

$$\text{wave velocity } v = \frac{\omega}{k} \Rightarrow \frac{v_{\max}}{v} = \frac{\omega a}{\omega/k} = ka.$$

From the given equation $k = \text{Co-efficient of } x = 6 \text{ m}^{-1}$

$$\Rightarrow \frac{v_{\max}}{v} = ka = 6 \times 60 \times 10^{-6} = 3.6 \times 10^{-4}$$

$$\text{here } a = 60 \text{ micron} = 60 \times 10^{-6} \text{ m}$$

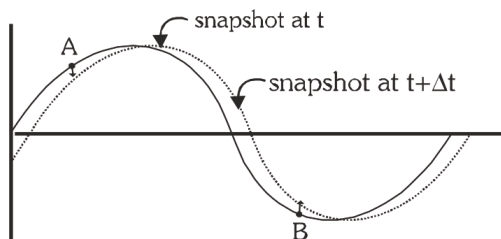
3. Ans (1)

$$v_{\text{Particle}} = -(\text{slope}) (v_{\text{wave}})$$

At A : slope $\rightarrow +ve$ so v_p is $-ve$.

At B : slope $\rightarrow -ve$ so v_p is $+ve$.

OR



4. Ans (1)

$$V = \frac{\lambda}{T} = \frac{0.5}{0.04} = \frac{25}{2} \text{ m/s}$$

$$V = \sqrt{\frac{T}{\mu}} \Rightarrow T = \mu v^2 = (0.04)(12.5)^2 = 6.25 \text{ N}$$

5. Ans (4)

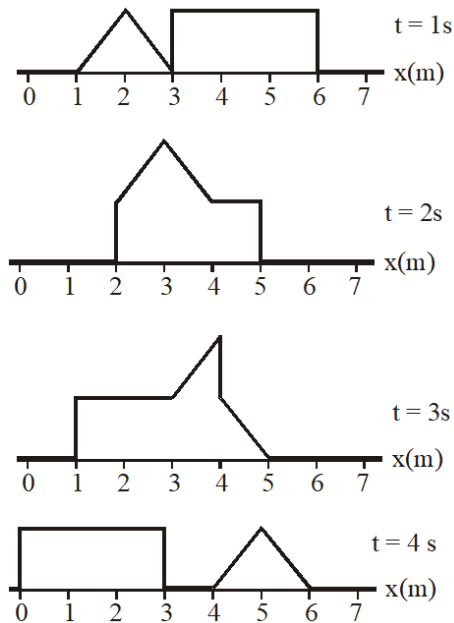
$$a_1 = 5 \quad \omega_1 = 150\pi, \quad n_1 = 75$$

$$a_2 = 10 \quad \omega_2 = 300\pi, \quad n_2 = 150$$

$$\frac{I_1}{I_2} = \left(\frac{a_1}{a_2}\right)^2 \left(\frac{n_1}{n_2}\right)^2$$

$$\frac{I_1}{I_2} = \left(\frac{5}{10}\right)^2 \left(\frac{75}{150}\right)^2 = \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^2 = \frac{1}{16}$$

6. Ans (3)



7. Ans (1)

When two waves having same amplitude and wavelength moving in opposite direction are superposed to each other then stationary wave will be produced.

8. Ans (2)

$$V_s = \sqrt{\frac{\gamma R T_1}{M_w}} \Rightarrow V_s \propto \sqrt{T}$$

$$\Rightarrow \frac{V_{s1}}{V_{s2}} = \sqrt{\frac{T_1}{T_2}}$$

$$\Rightarrow \frac{v}{\sqrt{3}v} = \sqrt{\frac{T}{T+600}}$$

$$T = 300 \text{ K} = 27^\circ\text{C}$$

9. Ans (2)

$$L_2 - L_1 = 20 \log_{10} \left(\frac{r_1}{r_2} \right)$$

$$80 - L_1 = 20 \log_{10} \left(\frac{400}{20} \right)$$

$$80 - L_1 = 20 \log_{10} 20$$

$$80 - L_1 = 20 \log_{10} (10 \times 2)$$

$$80 - L_1 = 20 [\log_{10} 10 + \log_{10} 2]$$

$$80 - L_1 = 20 [1 + 0.3]$$

$$80 - L_1 = 26$$

$$L_1 = 54 \text{ dB}$$

11. Ans (2)

$$[A] = \frac{[U][X + B^2]}{[\sqrt{X}]} = \frac{[ML^2T^{-2}][L]}{L^{1/2}} = [ML^{\frac{5}{2}}T^{-2}]$$

$$[B^2] = [L]$$

$$[AB^2] = [ML^{\frac{7}{2}}T^{-2}]$$

13. Ans (1)

$$n_1 u_1 = n_2 u_2$$

$$n_1 = n_2 = g$$

$$\frac{L_1}{L_2} = \left(\frac{T_1}{T_2} \right)^2 = 4$$

14. Ans (3)

$$q = ne$$

$$80 \times 10^{-6} = n \times 1.6 \times 10^{-19}$$

19. Ans (2)

$$\text{Given, } A = 0.3 \text{ m}^2, n = 2 \times 10^{25} / \text{m}^3,$$

$$q = 3t^2 + 5t + 2C$$

$$\therefore I = \frac{dq}{dt} = \frac{d}{dt} (3t^2 + 5t + 2) = 6t + 5$$

$$\text{At } t = 2 \text{ \& } I = 6 \times 2 + 5 = 17$$

$$\therefore \text{Drift velocity, } v_d = \frac{I}{neA}$$

$$= \frac{17}{2 \times 10^{25} \times 1.6 \times 10^{-19} \times 0.3}$$

$$= \frac{17}{0.96 \times 10^{-6}} = 1.77 \times 10^{-5} \text{ m/s}$$

20. Ans (2)

$$R = \frac{\rho \ell^2}{V} \Rightarrow \ell = \sqrt{\frac{RV}{\rho}} \rightarrow \ell = \sqrt{\frac{3 \times 3}{\rho}} = \frac{3}{\sqrt{\rho}} \text{ m}$$

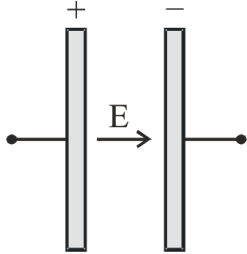
22. Ans (3)

$$\text{Charge on } 4\mu\text{F} = 4\mu\text{F} \times 6\text{V} = 24\mu\text{C} = Q_2$$

$$\text{Charge on } 5\mu\text{F} = \frac{3}{10} \times 6 \times 5 = 9\mu\text{C} = Q_1$$

$$\frac{Q_1}{Q_2} = \frac{9\mu\text{C}}{24\mu\text{C}} = \frac{3}{8}$$

23. Ans (2)



$$\text{Energy density } u = \frac{1}{2} \epsilon_0 E^2$$

$$\text{Volume of capacitor } V' = Ad$$

$$\text{Energy stored in capacitor}$$

$$U = uV' = \frac{1}{2} \epsilon_0 E^2 Ad$$

OR

$$U = \frac{1}{2} CV^2 = \frac{1}{2} \frac{A\epsilon_0}{d} V^2$$

$$= \frac{1}{2} \frac{A\epsilon_0}{d} (E/d)^2 = \frac{1}{2} \epsilon_0 E^2 Ad$$

24. Ans (3)

$$V_1 = \frac{kQ}{R} = 1600 \text{ volt}$$

$$\frac{4}{3} \pi R^3 = n \frac{4}{3} \pi r^3 \Rightarrow R = n^{1/3} r$$

$$Q = nq$$

$$V_2 = \frac{kq}{r} = k \left(\frac{Q}{n} \right) \frac{n^{1/3}}{R} = \frac{1}{n^{2/3}} \frac{kQ}{R}$$

$$\text{Here } n = 64, \quad n^{2/3} = 16$$

$$V_2 = \frac{1}{16} \quad V_1 = \frac{1600}{16} = 100 \text{ V}$$

27. Ans (2)

$$\frac{1}{2} \epsilon_0 E^2 = 2.1 \times 10^{-3}$$

$$\Rightarrow E = 21.6 \text{ N/C}$$

29. Ans (3)

Q-same

$$U' \propto \frac{1}{C}$$

$$C' = KC$$

$$U' = \frac{C}{C'} U_0 = \frac{U_0}{K} \frac{U_0}{K}$$

30. Ans (2)

In series combination Q is constant, hence according to

$$U = \frac{Q^2}{2C} \Rightarrow U \propto \frac{1}{C} \Rightarrow \frac{U_1}{U_2} = \frac{C_2}{C_1} = \frac{0.6}{0.3} = \frac{2}{1}$$

31. Ans (4)

$$U = \frac{Q^2}{2C}$$

$$\therefore C = \frac{\epsilon_0 A}{d}$$

$$\text{If } d \downarrow \quad C \uparrow \quad U \downarrow$$

SECTION-B

36. Ans (2)

$$a_R = \sqrt{a_1^2 + a_2^2 + 2a_1 a_2 \cos(\Delta\phi)}$$

$$\text{Here } a_1 = a_2 = a_R = a$$

$$\text{So } a = \sqrt{a^2 + a^2 + 2a^2 \cos(\Delta\phi)}$$

$$\cos(\Delta\phi) = -\frac{1}{2} \Rightarrow \Delta\phi = \frac{2\pi}{3}$$

37. Ans (2)

Tension in the middle and the top of rope are respectively

$$T_1 = mg/2 \text{ and } T_2 = mg$$

The ratio of speed of wave is

$$\frac{V_1}{V_2} = \frac{\sqrt{T_1/\mu}}{\sqrt{T_2/\mu}} = \sqrt{\frac{mg/2}{mg}} = \frac{1}{\sqrt{2}}$$

The ratio of wavelengths

$$\frac{\lambda_1}{\lambda_2} = \frac{V_1/f}{V_2/f} = \frac{V_1}{V_2} = \frac{1}{\sqrt{2}}$$

$$\therefore \lambda_2 = \sqrt{2}\lambda_1 = \sqrt{2}\lambda_0$$

38. Ans (3)

$$(1) \text{ same phase } I_R = 4I$$

$$(2) \phi = 90^\circ, I_R = I + I + 2\sqrt{I}\sqrt{I} \cos 90^\circ = 2I$$

$$\frac{I_1}{I_2} = \frac{4I}{2I} = 2 : 1$$

39. Ans (2)

Here $DL_1 = 40 \text{ m}$ and

$$DL_2 = \sqrt{40^2 + 9^2} = 41 \text{ m}$$

$$\text{Path difference } \Delta x = 41 - 40 = 1 \text{ m}$$

$$\text{For first maximum } \Delta x = \lambda = \frac{V}{f}$$

$$f = \frac{V}{\Delta x} = \frac{330}{1} = 330 \text{ Hz}$$

40. Ans (2)

$P^x Q^y c^z$ is dimensionless

$$[ML^{-1}T^{-2}]^x [MT^{-3}]^y [LT^{-1}]^z \propto M^0 L^0 T^0$$

$$M^{x+y} L^{-x+z} T^{-2x-3y-z} \propto M^0 L^0 T^0$$

Comparing powers of M, L and T

$$x + y = 0 \Rightarrow y = -x$$

$$-x + z = 0 \Rightarrow z = x$$

$$-2x - 3y - z = 0$$

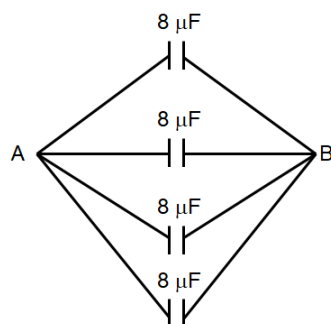
x and y are equal and of opposite sign

x and z are equal and of same sign

$$x = 1, y = -1, z = 1$$

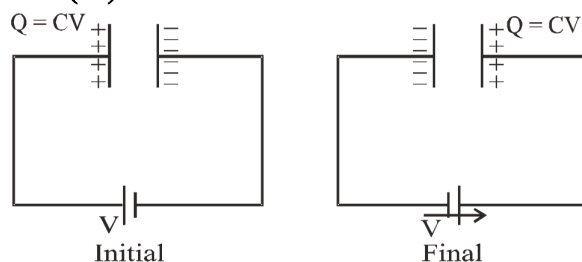
41. Ans (1)

Given circuit can be draw as



Equivalent capacitance = $4 \times 8 = 32 \mu F$

44. Ans (2)



Total charge flown through battery $Q = 2CV$

$\therefore$ Energy supplied by battery = QV

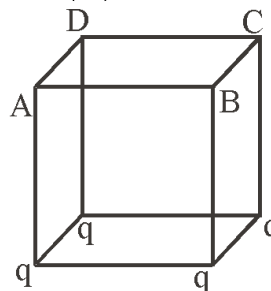
$$\Rightarrow 2 CVV$$

$$\Rightarrow 2 CV^2$$

$$\text{Heat loss} \Rightarrow 2 CV^2$$

Hence incorrect option is (2)

46. Ans (3)



Flux due to one charge

$$\phi = \frac{q}{24\epsilon_0}$$

due to four charge

$$\phi_{\text{net}} = \frac{q}{6\epsilon_0}$$

49. Ans (3)

Force on plate A = $Q_A E_B$

$$\text{Force} = Q \left(\frac{\sigma}{2\epsilon_0} \right) = \frac{Q^2}{2\epsilon_0 A}$$

At equilibrium

Force = kx

$$x = \frac{\text{Force}}{k} = \frac{Q^2}{2\epsilon_0 AK}$$

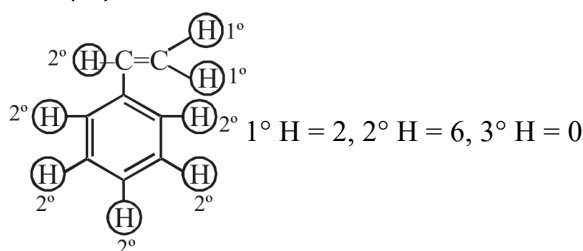
SUBJECT : CHEMISTRY

SECTION-A

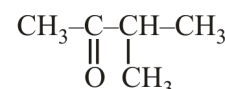
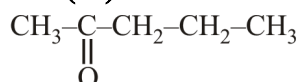
51. Ans (2)

NCERT Pg. # 319

56. Ans (2)



63. Ans (2)



64. Ans (3)

C N $n = 2$

Si P $n = 3$

EA 3^{rd} Pd. > EA 2^{nd} Pd.

$\Rightarrow \text{Si} > \text{C} > \text{P} > \text{N}$

stable configuration

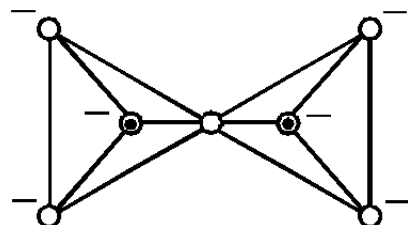
67. **Ans (4)**

Z_{eff}

${}_8\text{O} = 3.9$

${}_9\text{F} = 4.55$

70. **Ans (4)**



71. **Ans (3)**

CO neutral (neither acid nor base)

85. **Ans (4)**

The atomic radius of P is longest out of O,S,N and P. Therefore, P–H bond will be the longest one.

SECTION-B

92. **Ans (2)**

Oxidation no. $\propto$ acidic nature

99. **Ans (4)**

Z_1 = second ionisation potential and Z_2 = Third ionisation potential. Second ionisation potential is always less than the third ionisation potential.

100. **Ans (4)**

Isotopes have same number of protons (i.e. same atomic number). So they occupy same position in the periodic table. However, due to different numbers of neutrons their atomic weights are different.

SUBJECT : BIOLOGY-I

SECTION-A

102. **Ans (4)**

NCERT Pg. # 75

103. **Ans (4)**

NCERT Pg. # 75

105. **Ans (2)**

NCERT XII Pg. # 89

114. **Ans (4)**

NCERT (XII) Pg. # 79

119. **Ans (4)**

NCERT-XI, Pg. # 70

120. **Ans (2)**

NCERT-XI, Pg. # 80, 81

121. **Ans (4)**

NCERT XI # 79

122. **Ans (4)**

NCERT-XI, Pg. # 75

123. **Ans (3)**

NCERT (XI) Pg. # 71, 5.3.3

124. **Ans (3)**

NCERT-XI, Pg. # 74

125. **Ans (1)**

NCERT-XI, Pg. # 73

126. **Ans (3)**

NCERT-XII, Pg. # 278

127. **Ans (4)**

NCERT-XII, Pg # 281

128. **Ans (2)**

NCERT-XII, Pg. # 272

130. **Ans (1)**

NCERT-XII, Pg. # 272

131. **Ans (3)**

NCERT-XII, Pg. # 276, 281, 282

134. **Ans (4)**

NCERT-XII, Pg. # 283

135. **Ans (4)**

NCERT-XII, Pg. # 284, 267, 283

SECTION-B

136. **Ans (3)**

NCERT(XIIth) # P.No.118, 119

137. **Ans (1)**
NCERT (XII) Pg. # 118-120
138. **Ans (4)**
NCERT (XII) Pg. # 119
139. **Ans (1)**
NCERT (XIIth) Pg. # 73
143. **Ans (3)**
NCERT Pg.# 72, 76, 79, 80
147. **Ans (2)**
NCERT-XII, Pg. # 281, Fig. 16.7
149. **Ans (2)**
NCERT-XII, Pg. # 274

SUBJECT : BIOLOGY-II
SECTION-A

152. **Ans (1)**
NCERT-XII, Pg. # 270, 272, 274
153. **Ans (1)**
NCERT-XII, Pg. # 271
155. **Ans (3)**
NCERT-XII, Pg. # 261
156. **Ans (3)**
NCERT Pg. # 259, 260, 261
157. **Ans (1)**
NCERT-XII, Pg. # 266, 267
159. **Ans (2)**
NCERT-XII, Pg. # 288
160. **Ans (3)**
NCERT-XII, Pg. # 284, 285
161. **Ans (4)**
NCERT-XII, Pg. # 229
162. **Ans (2)**
NCERT-XII, Pg. # 232
163. **Ans (3)**
NCERT-XII, Pg. # 212

166. **Ans (4)**
NCERT (XIIth) Pg. # 29
167. **Ans (3)**
NCERT (XIIth) Pg. # 21
168. **Ans (4)**
NCERT (XII) Pg. # 25 (E), 27 (H)
170. **Ans (4)**
NCERT XIIth Pg.# 34
172. **Ans (2)**
NCERT XII, Page # 28
173. **Ans (2)**
NCERT Pg # 23
174. **Ans (3)**
NCERT Pg # 31
175. **Ans (4)**
NCERT Pg # 23
176. **Ans (2)**
NCERT Pg # 39
177. **Ans (4)**
NCERT Pg # 24
178. **Ans (4)**
NCERT Pg # 24
181. **Ans (2)**
NCERT Pg # 38
182. **Ans (3)**
NCERT Pg # 26,27
183. **Ans (3)**
NCERT Pg # 29
184. **Ans (2)**
NCERT Pg # 25
185. **Ans (1)**
NCERT Pg # 28

SECTION-B

186. **Ans (3)**
NCERT-XII, Pg. # 260, Fig.15.1

187. Ans (2)

NCERT-XII, Pg. # 260

189. Ans (1)

NCERT XI (E) Page 210, 211 and 213 (2nd para)

हिन्दी पेज # 210, 211, 212 (पहला पैराग्राफ) ,oa 213

(नीचे से दूसरा पैराग्राफ)

190. Ans (1)

NCERT (XIIth) Pg. # 209

193. Ans (2)

NCERT (XII) Pg. # 25,26

194. Ans (2)

NCERT (XII) Pg # 23

195. Ans (4)

NCERT XII, Page # 28, 29, 30, 31

197. Ans (2)

NCERT XII (E), Pg # 39, para-2, line 7,8,9,10

198. Ans (4)

NCERT Pg # 25

199. Ans (3)

NCERT Pg # 35

200. Ans (4)

NCERT Pg # 23